



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(10x3=30)

1. What is Yates correction for continuity?
2. Write down contingency table for two attributes A & B. What are the criteria for their independence?
3. What is meant by Fisher's least significant difference test and how is it used in interpreting the result of an experiment?
4. Five fertilizers A, B, C, D and E were tested in 5x5 Latin square design. Following information is given:

Source of Variation	Sum of Square	Degree of Freedom	Mean Square	F-ratio
Fertilizer			49.15	
Rows	46.67			
Columns			3.51	
Error				
Total			11.8892	

Complete the analysis by filling in the blanks.

5. What are the basic principles of experimental design?
6. What is meant by analysis of variance?
7. What is meant by partial correlation?
8. Calculate the multiple correlations $R_{2.12}$.
9. Describe the multiple linear regressions with two regressors.
10. In a trivariate distribution it is found that:

$$b_{12} = -0.10 \quad b_{13} = 0.27 \quad b_{23} = 0.67 \quad b_{21} = -0.40 \quad b_{31} = 0.60 \quad b_{32} = 0.38$$

$$s_1 = 1.0, \quad s_2 = 0.8, \quad s_3 = 9.0, \quad r_{12} = 0.6, \quad r_{23} = 0.65, \quad r_{31} = 0.7$$

Find $b_{23.1}$.

Answer the following questions

(5x6=30)

Q2. A survey of 200 families known to be regular television viewers was undertaken. They were asked which of 3 television channels they watch most during an average week. A summary of their replies is given in the following table together with the region in which they lived:

Channel	Region			
	North	East	South	West
PTV 1	29	16	42	23
PTV 2	6	11	26	7
STN	15	3	12	10

Test the hypothesis there is no association between the channel watched most and the region.

Q3. Given the following information:

	Sample			
	1	2	3	4
Sample size	4	6	7	3
Sample mean	58	57	43	42
Sample biased variance	10	30	6	9

Construct the analysis of variance table and test the hypothesis that population means are equal. Use 5% level of significance.

Q4. An industrial engineer is investigating the effects of four assembly methods A, B, C and D on the assembly time for color television components. 4 operators are selected for the study. The engineer Latin square design:

Order of Assembly	Operator			
	1	2	3	4
1	C 10	D 14	A 7	B 9
2	B 7	C 16	D 11	A 11
3	A 5	B 10	C 6	D 14
4	D 8	A 10	B 11	C 11

Analyze the data in the following table and draw appropriate conclusion for assembly methods.

Q5. Estimate a regression line from the following data of heights x and weight w of 12 person:

Height (x)	Weight (w)			
60	110	135	120	
62	120	140	130	135
64	150	145		
70	170	185	160	

Test the hypothesis that population regression coefficient $\beta = 0$.

Q6. Suppose that yield observation were taken at 7 different agricultural experimental stations across the country along with the rainfall and amount of fertilizer used. The following data is obtained:

Fertilizer x_1	1	2	3	4	5	6	7
Rainfall x_2	1	2	1	3	2	2	3
Yield y	4	5	5	7	6	7	8

Fit a multiple linear regression y on x_1 and x_2 .